Lecture 4

# Many levels: the postulates in $\mathbb{C}^n$

**Objectives.** *Many systems have more than two readings. Their states are the rays of $\mathbb{C}^n$. This lecture carries the four postulates and the density matrix from $\mathbb{C}^2$ to $\mathbb{C}^n$, and finds two things that a qubit could not show. First, an eigenvalue may have an eigenspace larger than a line, and a reading of it then keeps the whole part of the state inside that eigenspace. Next, under a constant Hamiltonian with many energy levels, a mean can oscillate only at the differences of the levels divided by $\hbar$, the Bohr frequencies. The lecture also shows that, between measurements, the same probabilities and means follow if the observables change in time and the states do not.*

By the end of this lecture you will be able to:

- Find the probability of a degenerate reading and the state after it.
- Write the evolution of a constant Hamiltonian by its energy levels, find its Bohr frequencies, and decide whether a state is stationary.
- Find an observable in time and its equation of motion, for a constant Hamiltonian.
- Tell a degenerate reading from a finer reading recorded only in part, by an observable that does not commute with the finer one.

## 4.1 More than two readings

A Stern–Gerlach magnet splits a beam of silver atoms into two. Beams of some other atoms split into three or more, and Lecture 7 finds which numbers can occur. The energy of an atom has many possible readings too. We need systems with more than two readings.

In the theory of Lecture 1, an observable is a self-adjoint matrix and its readings are its eigenvalues. By the spectral theorem, a self-adjoint matrix with $n$ distinct eigenvalues has $n$ orthonormal eigenvectors, so it acts on a space of dimension at least $n$. A system whose states are the rays of $\mathbb{C}^n$ is called an *$n$-level system*. The qubit is the case $n = 2$.

Our example in this part of the course has $n = 3$. It has an orthonormal basis $e_+$, $e_0$, $e_-$, which we keep in this order, and the Hamiltonian

$$H = \varepsilon\big(|e_+\rangle\langle e_+| + |e_-\rangle\langle e_-|\big) = \begin{pmatrix} \varepsilon & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \varepsilon \end{pmatrix}, \qquad \varepsilon > 0. \tag{4.1}$$

Its energies are 0, read in the state $e_0$, and $\varepsilon$, read in every state of the plane spanned by $e_+$ and $e_-$. Lecture 7 explains the labels +, 0, −. The box *The physics* below describes a system with this Hamiltonian.

**The physics**

The nitrogen–vacancy (NV) centre is a defect in diamond: a nitrogen atom in place of a carbon atom, next to a site from which a carbon atom is missing. In its negatively charged form, $NV^-$, the defect has two unpaired electrons. In the defect's lowest energy state their spins form a three-level system, with basis states labelled $m_s = +1, 0, -1$. Smaller effects are neglected here: interactions with nearby nuclei, strain in the crystal and electric fields. With no magnetic field, its effective Hamiltonian is then (4.1). The states $m_s = \pm 1$ then have the same energy, higher than that of $m_s = 0$ by $\varepsilon = hD$, where $h = 2\pi\hbar$ and $D \approx 2.87$ GHz at room temperature. Green laser light makes the centre glow, more brightly from $m_s = 0$ than from $m_s = \pm 1$. Over many repetitions, the light tells 0 from $\pm 1$, and it cannot tell $+1$ from $-1$. The same light also pumps most centres into $m_s = 0$ within a few cycles, so the state after this reading is not the one Lüders' rule gives. Microwaves at 2.87 GHz take the centre out of $m_s = 0$, and the glow dims. This is the resonance of Lecture 2, at the frequency that § 4.3 calls a Bohr frequency.

## 4.2 The postulates in $\mathbb{C}^n$

**Postulate 4.1** (A system of $n$ levels)**.**

1. *States.* The states are described by unit vectors $\psi \in \mathbb{C}^n$, and two unit vectors describe the same state if and only if they differ by a phase. A state is a ray $[\psi]$.
2. *Observables.* An observable is a self-adjoint $n \times n$ matrix $A = \sum_a a\, E_a$, where $a$ runs over the distinct eigenvalues and $E_a$ is the orthogonal projection onto the whole eigenspace of $a$. Its possible readings are its eigenvalues.
3. *Measurement.* An ideal measurement of $A$ on $[\psi]$ reads $a$ with probability
$$\mathrm{Prob}(a) = \langle \psi, E_a \psi \rangle = \|E_a \psi\|^2$$
(Born's rule). If the reading is $a$, which requires $\mathrm{Prob}(a) > 0$, the state immediately afterwards is the ray
$$\left[\frac{E_a \psi}{\|E_a \psi\|}\right]$$
(Lüders' rule).
4. *Evolution.* Between measurements the state vector obeys
$$\mathrm{i}\hbar \frac{\mathrm{d}\psi_t}{\mathrm{d}t} = H\psi_t,$$
where the Hamiltonian $H$ is the observable whose readings are the energies.

For $n = 2$ these are the postulates of Lecture 1. The Hamiltonian may depend on time, as in Lecture 2. In this lecture it is constant. The proof of Lemma 1.11 did not use $n = 2$, so the evolution $U_t = e^{-\mathrm{i}tH/\hbar}$ is unitary. The definitions of an ensemble and of its density matrix did not use $n = 2$ either, and neither did the theorem that the statistics depend on the density matrix only (Theorem 3.4). So for a density matrix $\rho$ on $\mathbb{C}^n$, a reading $a$ has probability $\mathrm{Tr}(\rho E_a)$ and, if this is positive, leaves the density matrix
$$\frac{E_a \rho E_a}{\mathrm{Tr}(\rho E_a)}.$$

**Remark 4.2** (The space of states)**.** The ray $[\psi]$ consists of the unit vectors of the line $\mathbb{C}\psi$, and it fixes the orthogonal projection $|\psi\rangle\langle\psi|$ onto that line, since a phase does not change $|\psi\rangle\langle\psi|$. Conversely, the range of an orthogonal projection of rank 1 is a line, and the unit vectors of that line form one ray. So the states correspond one to one to the lines through 0 in $\mathbb{C}^n$, and to the orthogonal projections of rank 1. The set of lines through 0 in $\mathbb{C}^n$ is the *complex projective space* $\mathbb{CP}^{n-1}$. A unit vector has $2n$ real coordinates subject to one condition, and the phase takes away one more real parameter, so a state has $2n-2$ real parameters. For $n=2$ these are the angles $\theta$ and $\phi$ of (1.5), and Theorem 1.3 identifies $\mathbb{CP}^1$ with the Bloch sphere. The map $\psi \mapsto \mathbf{r}(\psi)$ from the unit sphere $S^3$ of $\mathbb{C}^2$ onto $S^2$ is the *Hopf map*.

What is new is the size of $E_a$. An eigenvalue is *degenerate* if its eigenspace has dimension at least 2. That dimension is the trace of $E_a$. On $\mathbb{C}^2$, only a multiple $c\mathbb{1}$ has a degenerate eigenvalue (Proposition 1.7), and its reading tells nothing. For the Hamiltonian (4.1), the energy $\varepsilon$ has the projection $E_\varepsilon = |e_+\rangle\langle e_+| + |e_-\rangle\langle e_-|$. Take the state

$$\psi = \alpha e_+ + \beta e_0 + \gamma e_-, \qquad \text{where} \qquad |\alpha|^2 + |\beta|^2 + |\gamma|^2 = 1.$$

The reading $\varepsilon$ has probability $|\alpha|^2 + |\gamma|^2$. If this is positive, the reading leaves the ray of

$$\frac{\alpha e_+ + \gamma e_-}{\sqrt{|\alpha|^2 + |\gamma|^2}}.$$

After the reading, the state lies in the plane of $e_+$ and $e_-$. Lüders' rule keeps both amplitudes $\alpha$ and $\gamma$, with their relative phase. *Linear Algebra*, Ex. 10.22, previews this rule. Nothing below depends on it.

## 4.3 Energy levels and evolution

The distinct energies $\varepsilon_k$ of a Hamiltonian are its *energy levels*, and a level is degenerate if its eigenvalue is. By the spectral decomposition (*Linear Algebra*, Thm 10.13),

$$H = \sum_k \varepsilon_k E_k, \qquad E_k E_l = \delta_{kl} E_k, \qquad \sum_k E_k = \mathbb{1}, \tag{4.2}$$

where $E_k$ is the projection onto the eigenspace of $\varepsilon_k$. In Lecture 1 the two energies of a qubit were written $E_\pm$. From now on $E$ always denotes a projection, and energies are written $\varepsilon$.

We need one fact about matrices that commute with $H$. It is used again in Lecture 5 and Lecture 6.

> **Lemma 4.3** (Commuting with a Hamiltonian)**.** *Let $H = \sum_k \varepsilon_k E_k$ be self-adjoint, as in (4.2). A matrix $A$ commutes with $H$ if and only if it commutes with every $E_k$. In that case $A$ maps each eigenspace of $H$ into itself, and $E_l A E_k = 0$ for $l \neq k$.*

*Proof.* If $A$ commutes with every $E_k$, then $A$ commutes with $H = \sum_k \varepsilon_k E_k$. Conversely, let $AH = HA$, and let $v$ be in the eigenspace of $\varepsilon_k$. Then $H(Av) = AHv = \varepsilon_k Av$, so $Av$ is in the same eigenspace. Hence $AE_k$ maps into the eigenspace of $\varepsilon_k$, which $E_l$ maps to zero for $l \neq k$. So $E_l A E_k = 0$ for $l \neq k$. Inserting $\mathbb{1} = \sum_l E_l$ on the left gives $AE_k = \sum_l E_l A E_k = E_k A E_k$. Since $H$ is self-adjoint, the adjoint $A^\dagger$ commutes with $H$ too, so in the same way $A^\dagger E_k = E_k A^\dagger E_k$. Taking adjoints gives $E_k A = E_k A E_k$. Hence $AE_k = E_k A$. ∎

**Theorem 4.4** (Evolution by the energy levels). *Let $H = \sum_k \varepsilon_k E_k$ be a constant Hamiltonian, as in (4.2). Let $\rho$ be a density matrix and $\rho_t = U_t \rho U_t^\dagger$.*

1. *The evolution is*
$$U_t = \sum_k e^{-\mathrm{i}t\varepsilon_k/\hbar} E_k. \tag{4.3}$$
2. *The energy statistics do not change:* $\operatorname{Tr}(\rho_t E_k) = \operatorname{Tr}(\rho E_k)$ *for every $k$ and every $t$.*
3. *For every observable $A$,*
$$\operatorname{Tr}(\rho_t A) = \sum_{k,l} e^{-\mathrm{i}(\varepsilon_k - \varepsilon_l)t/\hbar} \operatorname{Tr}(E_k \rho E_l A). \tag{4.4}$$
4. *The density matrix is* stationary, *$\rho_t = \rho$ for all $t$, if and only if $[H, \rho] = 0$, and if and only if $\rho = \sum_k E_k \rho E_k$. A pure state $[\psi]$ is stationary if and only if $\psi$ is an eigenvector of $H$.*

*Proof.*

1. This is the spectral decomposition of $e^{-\mathrm{i}tH/\hbar}$ (*Linear Algebra*, Thm 10.13, for the function $\lambda \mapsto e^{-\mathrm{i}t\lambda/\hbar}$).
2. By (4.3) and $E_k E_l = \delta_{kl} E_k$, each $E_k$ commutes with $U_t$. By the cyclicity of the trace, $\operatorname{Tr}(U_t \rho U_t^\dagger E_k) = \operatorname{Tr}(\rho U_t^\dagger E_k U_t) = \operatorname{Tr}(\rho E_k)$.
3. By (4.3) and its adjoint,
$$\rho_t = \sum_{k,l} e^{-\mathrm{i}t\varepsilon_k/\hbar} e^{\mathrm{i}t\varepsilon_l/\hbar} E_k \rho E_l.$$
Multiply by $A$ and take the trace.
4. First, if $\rho_t = \rho$ for all $t$, the derivative of $\rho_t$ at $t = 0$ vanishes, and by (3.7) $[H, \rho] = 0$. Next, if $[H, \rho] = 0$, then $E_l \rho E_k = 0$ for $l \neq k$ by Lemma 4.3, so $\rho = \sum_{k,l} E_k \rho E_l = \sum_k E_k \rho E_k$. Finally, if $\rho = \sum_k E_k \rho E_k$, only the terms with $k = l$ remain in the expansion of $\rho_t$ above, their phases cancel, and $\rho_t = \rho$.
For a pure state $\rho = |\psi\rangle\langle\psi|$, suppose $H\rho = \rho H$. Applied to $\psi$, this gives $H\psi = |\psi\rangle\langle\psi| H\psi = \langle\psi, H\psi\rangle\, \psi$, so $\psi$ is an eigenvector. Conversely, if $H\psi = \varepsilon\psi$, then $H|\psi\rangle\langle\psi| = \varepsilon |\psi\rangle\langle\psi| = |\psi\rangle\langle\psi| H$, since $\varepsilon$ is real. ∎

*Linear Algebra*, Ex. 10.24, previews the formula (4.3). What the theorem adds is (4.4). The energies themselves never appear in a mean. Only their differences do, through the angular frequencies
$$\frac{\varepsilon_k - \varepsilon_l}{\hbar},$$
the *Bohr frequencies* of the system. They include 0, for $k = l$, and come in pairs of opposite sign. A mean can oscillate only at these frequencies, which are the same for every observable and every state. The observable and the state decide how much each frequency contributes, through the numbers $\operatorname{Tr}(E_k \rho E_l A)$, and some may contribute nothing. For example, a stationary state, or the observable $\mathbb{1}$, gives a constant mean. A qubit with energies $c \pm \frac{1}{2}\hbar\nu_0$ has the single positive Bohr frequency $\nu_0$. In Lecture 2, $\nu_0$ was the frequency of the resonance. Atoms absorb and emit light at their Bohr frequencies. This is Bohr's rule for spectral lines, named here and not derived.

Three warnings follow from the theorem. First, every nonzero vector of a degenerate eigenspace is an eigenvector of $H$, so every state of that eigenspace is stationary. For (4.1), the states $e_+$, $e_-$ and $\chi_+ = \frac{1}{\sqrt{2}}(e_+ + e_-)$ are all stationary. So the stationary pure states do not form a basis fixed by $H$ alone. Second, the energy statistics of every state are constant, and

yet a state may change in time. § 4.6 shows one. Third, the formula (4.3) needs $H$ constant. A drive that changes $H$ in time, as in Lecture 2, has its own evolution $U(t,s)$, which we construct for each drive we use.

## 4.4 Observables in time

So far the state has changed in time and the observables have stayed fixed. The same probabilities and means come out if, instead, the observables change in time and the state stays fixed. For a constant Hamiltonian $H$ and an observable $A$, the *observable at the time $t$* is

$$A_t = U_t^\dagger A U_t.$$

**Proposition 4.5** (Observables in time). *Let $H$ be a constant Hamiltonian, $A = \sum_a a\,E_a$ an observable, $\rho$ a density matrix and $\rho_t = U_t \rho U_t^\dagger$.*

1. $\operatorname{Tr}(\rho_t E_a) = \operatorname{Tr}\big(\rho\, U_t^\dagger E_a U_t\big)$ *for every reading $a$, and* $\operatorname{Tr}(\rho_t A) = \operatorname{Tr}(\rho A_t)$.
2. *The observable $A_t$ obeys the* equation of motion

$$\frac{\mathrm{d}A_t}{\mathrm{d}t} = \frac{\mathrm{i}}{\hbar}\,[H, A_t]\,,$$

*and $A_t = A$ for every $t$ if and only if $[A,H] = 0$.*

*Proof.* The first item is the cyclicity of the trace, $\operatorname{Tr}(U_t\rho U_t^\dagger E_a) = \operatorname{Tr}(\rho U_t^\dagger E_a U_t)$, multiplied by $a$ and added over $a$. For the second item, differentiating (4.3) term by term gives $\mathrm{d}U_t/\mathrm{d}t = -\frac{\mathrm{i}}{\hbar} H U_t$, and its adjoint gives $\mathrm{d}U_t^\dagger/\mathrm{d}t = \frac{\mathrm{i}}{\hbar} U_t^\dagger H$. By the product rule, and since $U_t$ commutes with $H$,

$$\frac{\mathrm{d}A_t}{\mathrm{d}t} = \frac{\mathrm{i}}{\hbar} U_t^\dagger (HA - AH) U_t = \frac{\mathrm{i}}{\hbar}(HA_t - A_t H).$$

If $[A,H] = 0$, then $A$ commutes with every $E_k$ by Lemma 4.3, hence with $U_t$ by (4.3), and $A_t = A$. Conversely, if $A_t = A$ for every $t$, the derivative at $t = 0$, $\frac{\mathrm{i}}{\hbar}[H,A]$, vanishes. ∎

Letting the observables change in time is the *Heisenberg picture*, and letting the states change is the *Schrödinger picture*. Between measurements they give the same probabilities and means. In this course the state changes in time, and Lüders' rule updates the state. Taking the mean of the equation of motion $\mathrm{d}A_t/\mathrm{d}t = \frac{\mathrm{i}}{\hbar}[H, A_t]$ in a state gives (1.10).

For example, take a qubit with $H = \frac{\hbar\nu}{2}\sigma_z$, so that $U_t = e^{-\mathrm{i}\nu t\sigma_z/2}$. Since $\sigma_x\sigma_z = -\sigma_z\sigma_x$, we have $\sigma_x U_t = U_t^\dagger \sigma_x$. With $\sigma_z\sigma_x = \mathrm{i}\sigma_y$, this gives

$$(\sigma_x)_t = \big(U_t^\dagger\big)^2 \sigma_x = \big(\cos(\nu t)\mathbb{1} + \mathrm{i}\sin(\nu t)\,\sigma_z\big)\sigma_x = \cos(\nu t)\,\sigma_x - \sin(\nu t)\,\sigma_y.$$

The mean of $(\sigma_x)_t$ is $r_x\cos\nu t - r_y\sin\nu t$. This is the precession of Lecture 1, read from the observable.

## 4.5 A degenerate reading

Lüders' rule projects onto the whole eigenspace of the reading. Why not onto some vector in it? The question becomes precise when one reading is coarser than another.

Let $A = \sum_a a\,E_a$ be an observable and $f$ a real function. The observable

$$f(A) = \sum_a f(a)\,E_a$$

reads $f(a)$ in every state in which $A$ reads $a$ with certainty. Each eigenvalue $b$ of $f(A)$ has the projection

$$G_b = \sum_{a:\, f(a)=b} E_a,$$

which may be the sum of several $E_a$. A reading of $f(A)$ tells only the value of $f$. For example, let $Z = \operatorname{diag}(1, 0, -1)$ in the basis $e_+$, $e_0$, $e_-$. Then the Hamiltonian (4.1) is $H = \varepsilon Z^2$, a function of $Z$ with $f(\pm 1) = \varepsilon$ and $f(0) = 0$.

**Proposition 4.6** (A degenerate reading is not a reading half recorded). *Let $A = \sum_a a\, E_a$ be an observable, $f$ a real function, $b$ an eigenvalue of $f(A)$ with projection $G_b = \sum_{f(a)=b} E_a$, and $\rho$ a density matrix with $\operatorname{Tr}(\rho G_b) > 0$. Compare two experiments. In the first, $f(A)$ is measured and reads $b$. In the second, $A$ is measured, and only whether $f(a) = b$ is recorded.*

1. *In both, the reading $b$ occurs with probability $\operatorname{Tr}(\rho G_b)$.*
2. *Afterwards the density matrices are*

$$\rho_1 = \frac{G_b \rho G_b}{\operatorname{Tr}(\rho G_b)} \quad \text{and} \quad \rho_2 = \frac{1}{\operatorname{Tr}(\rho G_b)} \sum_{f(a)=b} E_a \rho E_a,$$

   *and $\rho_1 - \rho_2$ is the sum of the cross terms $E_a \rho E_{a'}/\operatorname{Tr}(\rho G_b)$ with $a \neq a'$ and $f(a) = f(a') = b$.*
3. *Every observable $C$ that commutes with $A$ has the same mean in both: $\operatorname{Tr}(\rho_1 C) = \operatorname{Tr}(\rho_2 C)$.*

*Proof.*

1. Each reading $a$ of $A$ has probability $\operatorname{Tr}(\rho E_a)$, and these add up to $\operatorname{Tr}(\rho G_b)$ over the $a$ with $f(a) = b$.
2. The first matrix is Lüders' rule for $f(A)$. For the second, the systems that read a given $a$ form an ensemble with density matrix $E_a \rho E_a/\operatorname{Tr}(\rho E_a)$, by Theorem 3.4. Pooling these branches, each weighted by its share $\operatorname{Tr}(\rho E_a)/\operatorname{Tr}(\rho G_b)$ of the systems, gives $\rho_2$, as in the derivation of (3.8). Expanding $G_b \rho G_b = \sum_{a,a'} E_a \rho E_{a'}$ over $f(a) = f(a') = b$ gives the difference.
3. By Lemma 4.3, applied to the self-adjoint matrix $A$, $C$ commutes with every $E_a$. So for $a \neq a'$, by the cyclicity of the trace, the commutation and $E_{a'} E_a = 0$,

$$\operatorname{Tr}(E_a \rho E_{a'} C) = \operatorname{Tr}(\rho E_{a'} C E_a) = \operatorname{Tr}(\rho C E_{a'} E_a) = 0. \qquad \blacksquare$$

In other words, a degenerate reading keeps the cross terms inside its eigenspace, and a finer reading that is recorded only in part loses them. Lüders' rule picks no vector in the eigenspace. It keeps the whole projection of the state. The two experiments agree on every observable that commutes with $A$. Only an observable that does not commute with $A$ can tell them apart. § 4.6 gives one.

**Remark 4.7** (Conditioning changes the weights, in $\mathbb{C}^3$). Remark 3.5 warned that after a branch an ensemble has new weights, and said that the error of keeping the old ones shows in $\mathbb{C}^n$. Take the ensemble

$$\left\{\left(\frac{1}{2}, e_+\right), \left(\frac{1}{2}, \frac{1}{\sqrt{2}}(e_0 + e_-)\right)\right\}$$

and measure the energy of (4.1). The reading $\varepsilon$ has probability $\frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4}$. The first member stays $e_+$, and the second goes to $e_-$. By Bayes' rule, each new weight is the

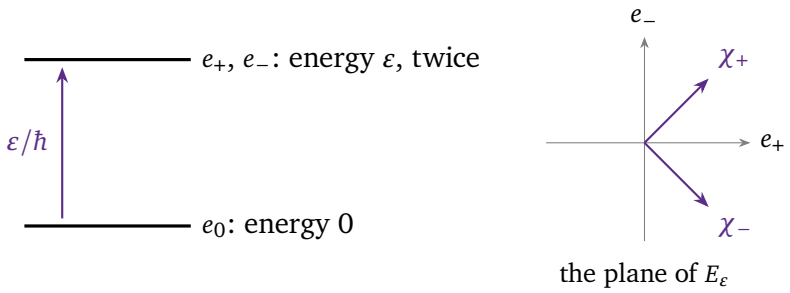



Figure 4.1: The three-level system (4.1). Left: the levels 0 and $\varepsilon$, and the one positive Bohr frequency $\varepsilon/\hbar$. Right: the plane of the degenerate level, with two of its orthonormal bases. The observable $Z$ reads $\pm 1$ on $e_\pm$, and $X$ reads $\pm 1$ on $\chi_\pm = \frac{1}{\sqrt{2}}(e_+ \pm e_-)$; both read 0 on $e_0$.

old weight times the probability of $\varepsilon$, divided by $\frac{3}{4}$. So the weights become $\frac{1}{2}/\frac{3}{4} = \frac{2}{3}$ and $\frac{1}{4}/\frac{3}{4} = \frac{1}{3}$, so a reading of $Z$ then gives $+1$ with probability $\frac{2}{3}$. The old weights would give $\frac{1}{2}$, a wrong prediction.

## 4.6 A three-level system

Figure 4.1 shows the system. Besides $Z = \operatorname{diag}(1, 0, -1)$ we use the observable

$$X = |e_+\rangle\langle e_-| + |e_-\rangle\langle e_+| = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix},$$

whose eigenvectors are $\chi_\pm = \frac{1}{\sqrt{2}}(e_+ \pm e_-)$, with the readings $\pm 1$, and $e_0$, with the reading 0. Both $Z$ and $X$ commute with $H$, and $XZ \neq ZX$. Lecture 5 studies these three observables, $H$, $Z$ and $X$, together.

**Example 4.8** (A degenerate reading keeps the relative phase)**.**

1. *Prepare.* The system starts in
$$\psi = \frac{1}{\sqrt{3}}(e_+ + e_0 + e_-).$$
A measurement of the energy reads 0 with probability $\frac{1}{3}$ and $\varepsilon$ with probability $\frac{2}{3}$. By Theorem 4.4, these probabilities are the same at every time, and the mean energy is always $\frac{2}{3}\varepsilon$.
2. *Evolve.* By (4.3),
$$\psi_t = \frac{1}{\sqrt{3}}\big(e_0 + e^{-i\varepsilon t/\hbar}(e_+ + e_-)\big).$$
A measurement of the projection $|\psi\rangle\langle\psi|$ asks whether the system is still in its initial state. It reads 1 with probability
$$|\langle\psi, \psi_t\rangle|^2 = \frac{1}{9}\left|1 + 2e^{-i\varepsilon t/\hbar}\right|^2 = \frac{5 + 4\cos(\varepsilon t/\hbar)}{9}.$$
The one positive Bohr frequency $\varepsilon/\hbar$ appears. The probability falls to $\frac{1}{9}$ at $t = \pi\hbar/\varepsilon$ and returns to 1 at $t = 2\pi\hbar/\varepsilon$. So the state $\psi$ changes in time, although its energy statistics do not.

3. *Read the energy at $t = \pi\hbar/\varepsilon$.* There $e^{-i\varepsilon t/\hbar} = -1$ and $\psi_t = \frac{1}{\sqrt{3}}(e_0 - e_+ - e_-)$. The reading $\varepsilon$ has probability $\frac{2}{3}$. By Lüders' rule the state afterwards is the ray of $-\frac{1}{\sqrt{2}}(e_+ + e_-)$, which is the ray of $\chi_+$. The reading has kept the relative phase of $e_+$ and $e_-$. The state $\chi_+$ is an eigenvector of $H$, so this state is stationary.
4. *Read $X$.* The state $\chi_+$ reads +1 with certainty.
5. *The same with $Z$, recorded in part.* Suppose that in step 3, instead of the energy, $Z$ had been measured and only "not 0" recorded. By Proposition 4.6, the reading "not 0" has the same probability $\frac{2}{3}$, and the state afterwards is

$$\frac{1}{2}\big(|e_+\rangle\langle e_+| + |e_-\rangle\langle e_-|\big),$$

since $\psi_t$ gives $e_+$ and $e_-$ equal weights. Then $X$ reads +1 with probability $\frac{1}{2}$, not 1. The observable $X$ does not commute with $Z$, and it tells the two experiments apart.

For the NV centre of § 4.1, $\varepsilon/\hbar = 2\pi D$, so the return time $2\pi\hbar/\varepsilon = 1/D$ is 0.348 ns.

**A system of $n$ levels**

*States:* the rays of $\mathbb{C}^n$, and the density matrices of ensembles. *Observables:* the self-adjoint $n \times n$ matrices $A = \sum_a a\,E_a$. An eigenvalue may be degenerate. *Evolution* under a constant $H = \sum_k \varepsilon_k E_k$: $U_t = \sum_k e^{-it\varepsilon_k/\hbar} E_k$. A mean can oscillate only at the Bohr frequencies $(\varepsilon_k - \varepsilon_l)/\hbar$, and $\rho$ is stationary if and only if $[H, \rho] = 0$ (Theorem 4.4). Equivalently, between measurements, the observables change in time, $A_t = U_t^\dagger A U_t$ (Proposition 4.5). *Measurement:* the reading $a$ has probability $\mathrm{Tr}(\rho E_a)$ and leaves $E_a \rho E_a / \mathrm{Tr}(\rho E_a)$, where $E_a$ projects onto the whole eigenspace of $a$.

**Summary.**

- *Evolution by the energy levels*: under a constant Hamiltonian, the energy statistics are constant, every mean is a sum of terms oscillating at the Bohr frequencies, and a state is stationary if and only if its density matrix commutes with $H$ (Theorem 4.4).
- *Observables in time*: for a constant Hamiltonian, $A_t = U_t^\dagger A U_t$ gives the same statistics between measurements, and $A_t = A$ for all $t$ exactly when $[A, H] = 0$ (Proposition 4.5).
- *A degenerate reading is not a reading half recorded*: both give the same probabilities, but the degenerate reading keeps the cross terms inside its eigenspace (Proposition 4.6).

Lecture 5 asks when two observables can be read together, starting with $Z$ and $X$.

## References

Cohen-Tannoudji, Diu and Laloë, *Quantum Mechanics*, vol. 1, Ch. III, §B-3, pp. 217–222, and §D-2, pp. 245–250; Complement $G_{III}$, pp. 317–319. Sakurai and Napolitano, *Modern Quantum Mechanics*, §2.1.3–2.1.4, pp. 67–69, and §2.2, pp. 75–82. Nielsen and Chuang, *Quantum Computation and Quantum Information*, §2.2.5, pp. 87–88. Doherty et al., "The nitrogen-vacancy colour centre in diamond", *Physics Reports* 528, 1 (2013).